PIR Motion Detection Alarm using Arduino Uno

Abstract

This project demonstrates a motion detection alarm system designed and simulated using Tinkercad Circuits. A PIR sensor detects movement, displays the system status on a 16x2 LCD, and activates a buzzer whenever motion is detected. The project introduces sensor interfacing, LCD communication, and digital input/output programming using Arduino Uno.

Objectives

- Learn PIR sensor interfacing with Arduino.
- Display motion status on a 16x2 LCD.
- Activate a buzzer during motion detection.
- Simulate and verify the circuit using Tinkercad.

Components Required

Arduino Uno, PIR Sensor, 16x2 LCD, Piezo Buzzer, Breadboard, Jumper Wires, Potentiometer (LCD Contrast), USB Power Supply.

Software Used

Tinkercad Circuits, Arduino IDE (Embedded C/C++).

Working Principle

The PIR sensor continuously monitors infrared radiation. When a moving object enters its detection area, the sensor output becomes HIGH. The Arduino reads this signal, displays 'Motion Detected!' on the LCD, and activates the buzzer for one second. When no movement is present, the LCD displays 'No Motion' and the buzzer remains OFF.

Pin Configuration

PIR OUT → D2
Buzzer → D3
LCD RS,E,D4,D5,D6,D7 → D7,D8,D9,D10,D11,D12

Applications

Home security systems, Automatic lighting, Intruder detection, Smart buildings, Office monitoring.

Learning Outcomes

Arduino programming, Digital input/output, PIR sensor interfacing, LCD interfacing, Embedded system design, Tinkercad simulation.

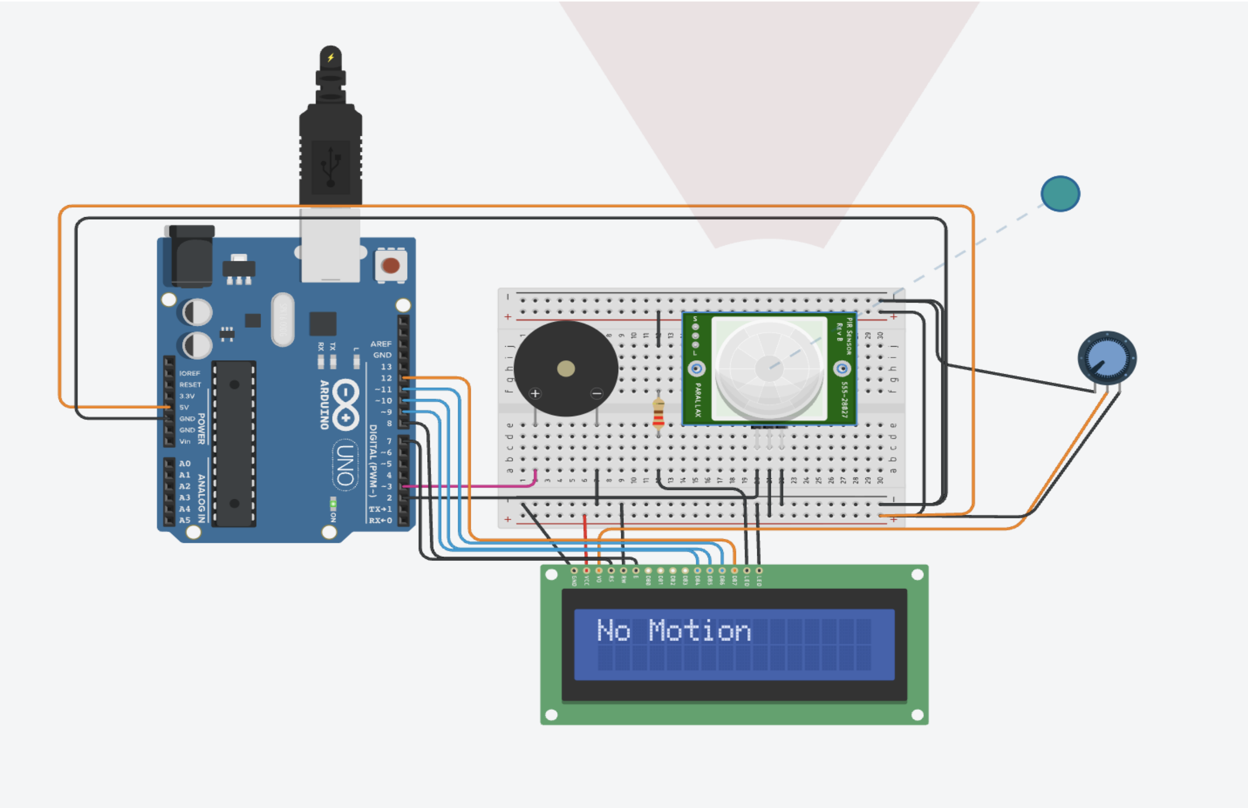
Conclusion

The project successfully detects motion using a PIR sensor and provides both visual and audible alerts. It demonstrates fundamental embedded system concepts and practical Arduino interfacing skills.

Arduino Source Code

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 8, 9, 10, 11, 12);
const int pirPin = 2;
const int buzzerPin = 3;
void setup() {
lcd.begin(16, 2);
lcd.print("PIR Sensor Init");
pinMode(pirPin, INPUT);
pinMode(buzzerPin, OUTPUT);
delay(5000);
lcd.clear();
lcd.print("Ready to Detect");
}
void loop() {
int motionDetected = digitalRead(pirPin);
if (motionDetected == HIGH) {
lcd.clear();
lcd.print("Motion Detected!");
digitalWrite(buzzerPin, HIGH);
delay(1000);
digitalWrite(buzzerPin, LOW);
} else
{
lcd.clear();
lcd.print("No Motion");
delay(500);
}
}
```

Simulation - No Motion



Simulation - Motion Detected